MANTHAN SHRIRAM JOSHI

Boston, MA | shriramjoshi.m@northeastern.edu | +1 617 735 5727 | LinkedIn | *Available for co-op: June 29 – December 13, 2026*

Education

Northeastern University, Boston, MA | *M.S. in Cell and Gene Therapies* GPA: 3.7/4 *December 2026*
Coursework: Immunology, Cell and Gene Therapies, Stem Cells and Regeneration, Advanced Drug Delivery Systems, Bioinformatics

Manipal Institute of Technology (MAHE), Manipal, India | *B.Tech in Biotechnology* | Minor: Pharmaceutical Biotechnology *July 2023*
Coursework: Genetics, Immunology, Bioinformatics, Pharmaceutical Science, Microbiology, Biochemistry, Bioprocess Engineering

Certifications

- **Harvard Business School** | Certificate in Management Essentials (Aug 2025): 8-week, 35-hour program in decision-making, implementation, organizational learning, and change management.
- **BioGrademy** | Good Clinical Practice (GCP) and Medical Writing Certifications.

Laboratory Skills

Cell Culture & Aseptic Technique	Aseptic handling and maintenance of mammalian cell lines (CHO) across 5+ passages at >80% viability; cryopreservation and recovery; drug-selection optimization (puromycin, 6-thioguanine); kill curve design using CellTiter-Glo 2.0 cytotoxicity assays; mammalian cell preparation for 3D bioink formulation.
Genome Engineering	End-to-end CRISPR/Cas9 knock-in and knock-out workflow design; Cas9-RNP complex assembly with HPRT1-targeting sgRNAs; CRISPRmax lipofection in 24-well format; transient transfection; lentiviral vector handling and titer determination.
Plasmid & Cloning	Bacterial transformation, colony screening, miniprep purification; restriction-enzyme digestion (NotI, EcoRV) for construct verification; validation of multiple plasmid constructs by gel readout.
PCR, Sequencing & Validation	Genomic DNA extraction; Phusion PCR amplification of target loci; primer design in Benchling for diagnostic amplicons (600–800 bp) and knock-in confirmation primers flanking homology arms; Sanger sequencing and ICE indel analysis; agarose gel electrophoresis; Nanodrop quantification.
Protein Biochemistry	Recombinant membrane protein expression and purification by Ni-NTA affinity chromatography; SDS-PAGE; Western blotting; Coomassie staining; ELISA; cytotoxicity assays.
Imaging & Software	Fluorescence microscopy for transgene confirmation; Benchling (primer design, construct mapping); GraphPad Prism (dose-response curve fitting); ICE web tool; Python and R (basic).

Experience

Northeastern University, *Teaching Assistant – Cell and Gene Therapies (Graduate Course)*, Boston, MA *Jan 2026 – Apr 2026*
- Reinforced graduate-level CGT concepts in office hours and discussion sessions, integrating recent literature on CAR-T therapy and iPSC-to-T-cell differentiation.
- Reviewed and provided structured written feedback on student project deliverables; conducted in-depth literature review of allogeneic and autologous CAR-T platforms to support classroom case discussions.

Northeastern University, *Graduate Research Training | CGT Lab*, Boston, MA *Sept 2025 – Dec 2025*
- Maintained CHO cell cultures across 5+ passages at >80% viability; assembled and delivered Cas9-RNP complexes with 2 HPRT1-targeting sgRNAs via CRISPRmax lipofection in 24-well plates for an end-to-end HPRT1 knockout/knock-in simulation.
- Ran CellTiter-Glo 2.0 cytotoxicity assays across 8 serial 3-fold dilutions of 6-thioguanine, graphed dose-response in GraphPad Prism, and selected 100 μM ($\sim$7% viability) to enrich HPRT1-KO cells; confirmed by fluorescence microscopy.
- Extracted gDNA from 6-TG-selected pellets, amplified HPRT1 by Phusion PCR, and submitted 8 amplicons for Sanger sequencing and ICE indel analysis to confirm gene disruption.
- Validated 5 plasmid constructs by NotI/EcoRV restriction digest and gel electrophoresis; designed 2 primer sets in Benchling (600–800 bp diagnostic amplicon and knock-in confirmation primers flanking homology arms).

Reagene Biosciences Pvt Ltd, *Research Intern*, Bangalore, India *Aug 2023 – Aug 2025*
- Contributed to 3D bioprinted human tissue model platforms developed as alternatives to animal testing for early-stage drug screening.
- Performed molecular biology and protein workflows: DNA purification, agarose gel electrophoresis, Nanodrop quantification, bacterial transformation, Western blotting, ELISA, and cytotoxicity assays.
- Maintained mammalian cell cultures and managed cryopreservation to deliver high-viability cell populations for bioink formulation.
- Supported business development with investor decks, grant proposals, and competitive market research on 3D bioprinting and non-animal testing platforms.

Tranalab Pvt Ltd / Centre for Human Genetics, *Undergraduate Research Intern (Thesis)*, Bangalore, India *Jan 2023 – Aug 2023*
- Thesis: *Enhancing Solubilization and Purification of M4, a Recombinant Membrane Protein Domain* – awarded first rank in 2019 batch. Optimized recombinant protein expression and purification including Ni-NTA affinity chromatography and SDS-PAGE for yield and purity assessment.
- Authored SOPs for protein expression, lysis, and purification used by downstream team members; maintained CHO cell cultures and cryogenic storage protocols supporting recombinant protein production.

Indian Institute of Technology, *Undergraduate Research Intern*, Indore, India *Sept 2021 – Oct 2021*
- Supported a stem-cell-based tissue growth and tumor control project through experimental design, MSDS preparation, and protocol drafting; assisted with target identification and therapeutic intervention modeling for early-phase concept work.

Leadership

Vice President, Graduate Biotech-Bioinfo Association, Northeastern University, Boston, MA *Dec 2024 – Jan 2026*
- Led executive-board operations including communications, records, and cross-committee coordination, supporting professional-development programming and member engagement.
- Planned and executed networking events, technical workshops, and industry panel discussions to expand community involvement and exposure to CGT and biotech employers.